

DATA SCIENCE PROJECT

Customer Segmentation for Targeted Retention and Value Growth

A robust RFM-based analysis of e-commerce purchase behaviour

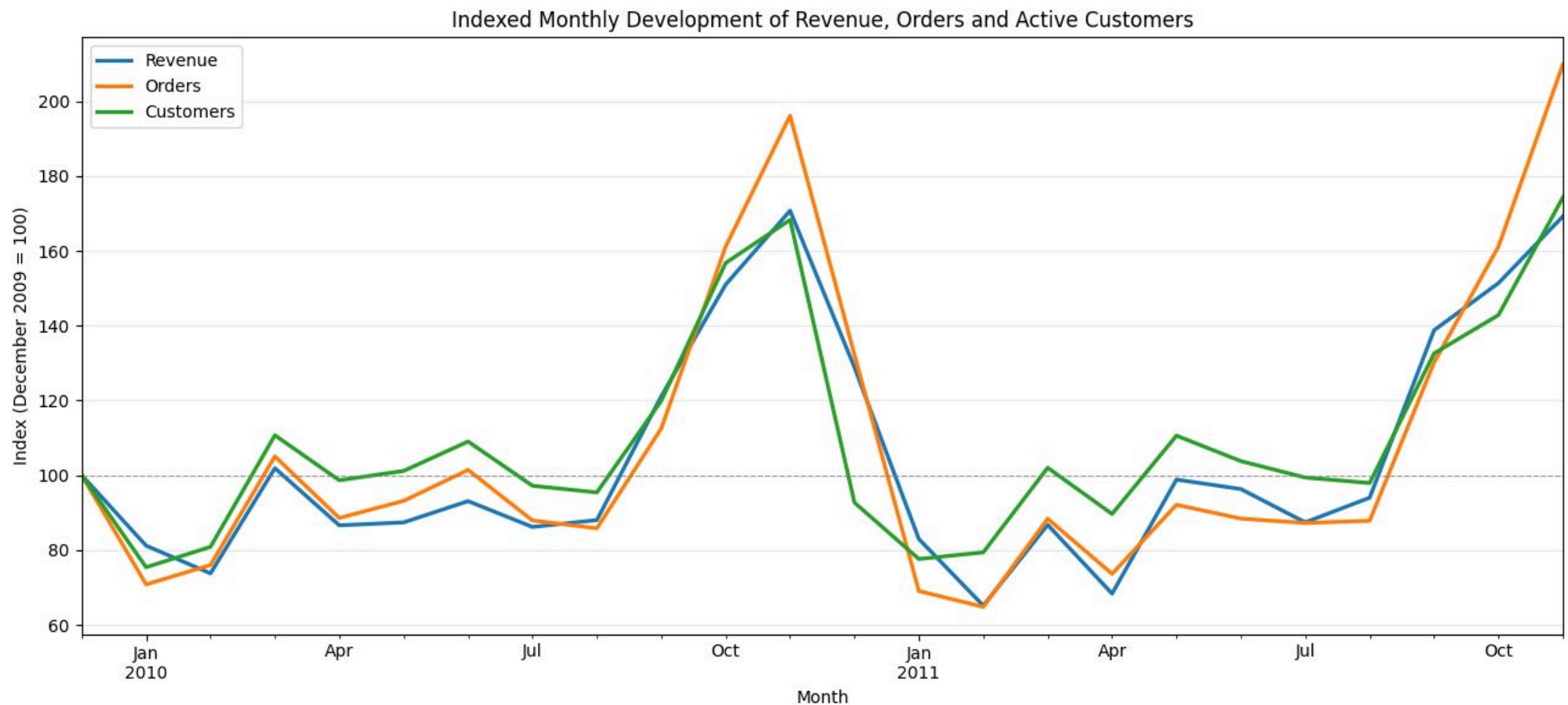
 **Presentation:** Heike Thöricht

 **Bremen**, 2 September 2026

Data source: UCI Online Retail II dataset (2009–2011); customer-level analysis based on 5,879 identifiable customers.

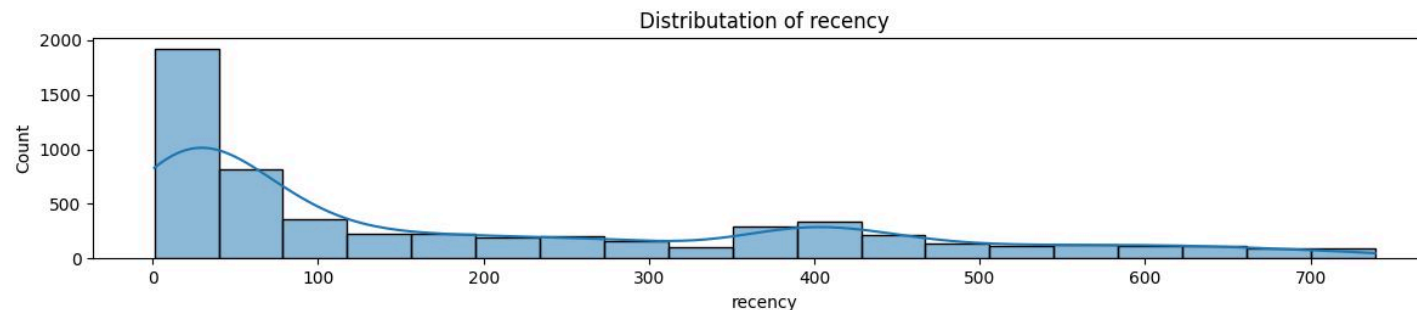
Customer value and engagement vary strongly

Most customers purchase infrequently and spend relatively little, while a small group generates disproportionately high order frequency and spend.



Why customer segmentation is needed

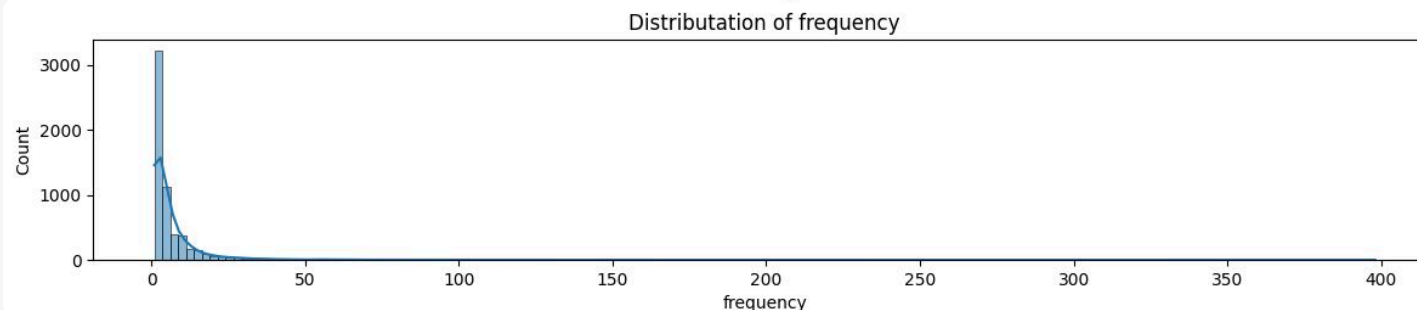
Distribution of customer behaviour across three key RFM dimensions — revealing how recently, how often, and how much each customer has purchased. Log transformation and p99 capping reduce the influence of extreme values before clustering.



Recency

Days since last purchase

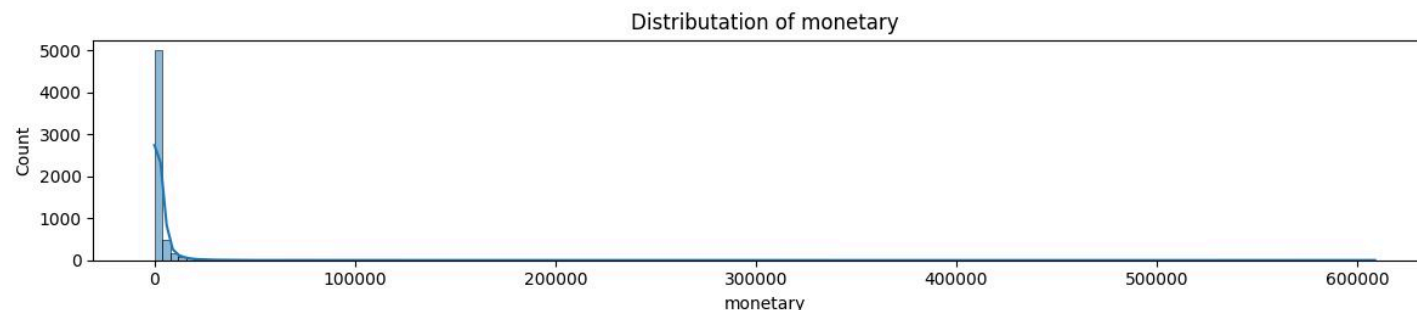
Right-skewed



Frequency

Number of purchases

Highly right-skewed



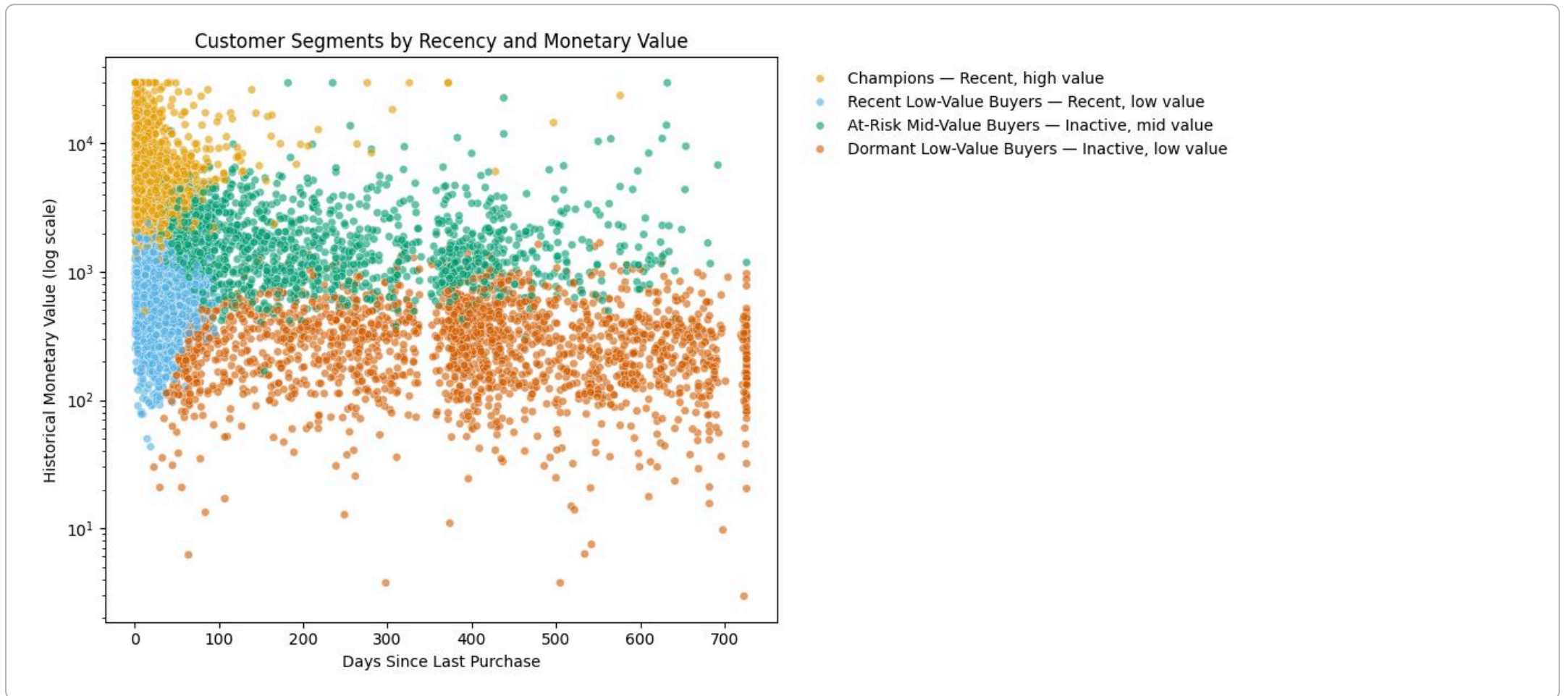
Monetary

Historical customer spend

Highly right-skewed

The observed heterogeneity indicates that customers should not be treated as one homogeneous group. We therefore use RFM-based clustering to identify actionable customer segments.

Four distinct customer segments emerge from RFM clustering



Segments are balanced, stable and actionable

The four segments differ clearly in purchase activity, repeat behaviour and customer value.

Customer segment	Customers	Share	Recency	Frequency	Monetary
Champions	1,219	20.8%	17 days	13 orders	5,067
Recent Low-Value Buyers	1,238	21.1%	24 days	3 orders	730
At-Risk Mid-Value Customers	1,457	24.8%	185 days	4 orders	1,496
Dormant Low-Value Customers	1,964	33.4%	405 days	1 order	282

Median values; n = 5,879 identifiable customers. Robust RFM clustering: silhouette score = 0.369; mean ARI = 0.980.

Segment-specific actions prioritise retention and customer value

The segmentation enables differentiated CRM actions instead of one-size-fits-all communication.

Champions

Goal: Retain and grow value

Action: VIP programme, cross-sell, personalised recommendations

Priority: HIGH

Recent Low-Value Buyers

Goal: Encourage repeat purchase

Action: Post-purchase journey, bundle or second-purchase incentive

Priority: MEDIUM-HIGH

At-Risk Mid-Value Customers

Goal: Win back historical value

Action: Personalised reactivation campaign

Priority: HIGH

Dormant Low-Value Customers

Goal: Limit cost of reactivation

Action: Low-cost automated outreach or suppression

Priority: LOW

- ❑ Priority focus: Retain Champions and reactivate At-risk customers; use scalable journeys to develop Recent low-value buyers and manage Dormant customers cost-effectively.

A robust RFM model provides the clearest customer segmentation

Adding NLP and behavioural features did not improve cluster separation — the Robust RFM model outperformed all extended variants.

Model	Best silhouette score	Assessment
RFM Baseline	0.439	Strong initial benchmark
Robust RFM	0.440	Best overall separation; selected core approach
Robust RFM + NLP	0.383	Text features did not improve segmentation
Robust RFM + Behaviour	0.209	Behaviour features reduced cluster separation

The four-cluster Robust RFM solution was selected for business use because it combined strong stability (mean ARI = 0.980; SD = 0.006), balanced segment sizes, and actionable customer profiles — with an acceptable silhouette score of 0.369.

Footnote: Best silhouette scores are shown across tested values of k . The selected final solution uses four clusters; $k = 2$ achieved the highest silhouette score but was too coarse for actionable segmentation.

Customer segments show distinct seasonal and timing patterns

Temporal features were used for post-clustering profiling and campaign activation rather than for defining the core segments.

Understanding when each segment is most active unlocks targeted campaign timing. Seasonal concentration and day-of-week patterns vary markedly across segments — for example, **Recent Low-Value buyers** are overwhelmingly autumn-driven (85.9%), while **Champions** skew towards weekday Thursday peaks. These patterns enable precise scheduling of reactivation, retention, and acquisition campaigns to reach each segment at the moments they are most engaged.

Segment	Autumn	Spring	Summer	Winter
At-Risk Mid-Value Customers	40.6%	26.4%	19.1%	13.9%
Champions	59.8%	13.9%	12.8%	13.5%
Dormant Low-Value Customers	36.6%	26.5%	18.6%	18.3%
Recent Low-Value Buyers	85.9%	4.8%	4.2%	5.2%

Segment	Mo	Di	Mi	Do	Fr	Sa	So
At-Risk Mid-Value Customers	24.2%	22.4%	17.8%	18.8%	7.6%	0.0%	9.1%
Champions	18.5%	20.3%	19.8%	22.9%	9.2%	0.0%	9.4%
Dormant Low-Value Customers	18.9%	19.5%	19.3%	18.4%	11.6%	0.1%	12.3%
Recent Low-Value Buyers	23.9%	22.6%	17.5%	17.0%	9.0%	0.0%	10.0%

Use the RFM segment as the strategic customer definition and use temporal profiles to optimise campaign timing.

ADD ON

Additional insights for sales and marketing

Insight	Evidence	Implication
Midday is the dominant preferred period	49.4–71.6% across segments	Test campaign delivery around 10:00–14:00
Weekdays dominate	Saturday share is 0.0–0.1%; Sunday 9.1–12.3%	Prioritise weekday communication
Beginning of month is most common	40.1–48.1% across segments	Concentrate selected promotions early in the month
Autumn is strongest	36.6–85.9% across segments	Prioritise seasonal campaigns in autumn

Key takeaways and recommendations

1 Robust RFM delivers the clearest segmentation

P99-capped frequency and monetary values provide the most stable and interpretable customer segments.

2 NLP & behavioral features are best for profiling

They do not improve cluster separation and should be reserved for segment activation and targeting refinement.

3 Four well-balanced, actionable segments

Champions, At-Risk Mid-Value, Dormant Low-Value, and Recent Low-Value buyers are manageable from a business perspective.

4 Highly stable solution

Stability analysis confirms robustness: mean ARI = 0.980 (SD = 0.006).

5 Temporal patterns enable targeted campaigns

Clear seasonal and day-of-week profiles can be leveraged for optimised campaign timing across all segments.

 **Recommendation:** Adopt the four-cluster Robust RFM model as the core customer segmentation framework and use temporal profiles for campaign activation and timing optimisation.